mounts to further reduce the remaining

omotive engine configuration. An even
l ignition has loosened that constraint
ize the firing sequence to minimize the
gines, which fire every piston stroke,
ments than four-stroke engines that fire

t of the crankshaft and the part of the
nk pin can be counterbalanced exactly
Therefore, only the balancing of the

th each piston is in the direction of the

$\psi_i)\Big]$, $i = 1, 2, \ldots, n$ (16.30)

ank radius, the connecting rod length,
angle θ gives the position of the piston
der axis is oriented at an angle of ψ_i
he direction of the x axis, the angle θ is
In most engines and compressors, $\overline{m}_{B_i}$,
subscript i must be maintained for the
e from crank 1 to crank i as shown in
es are usually included in the equations

forces from all of the cylinders. For the
n Figure 16.14. The z axis is along the
the piston travel of the first cylinder, and
em results. Then, the x component of the

$$\cos 2(\theta + \phi_i - \psi_i)\Big] \cos\psi_i \tag{16.31}$$

$$\cos 2(\theta + \phi_i - \psi_i)\Big] \sin\psi_i \tag{16.32}$$

masses is given by summing the vector

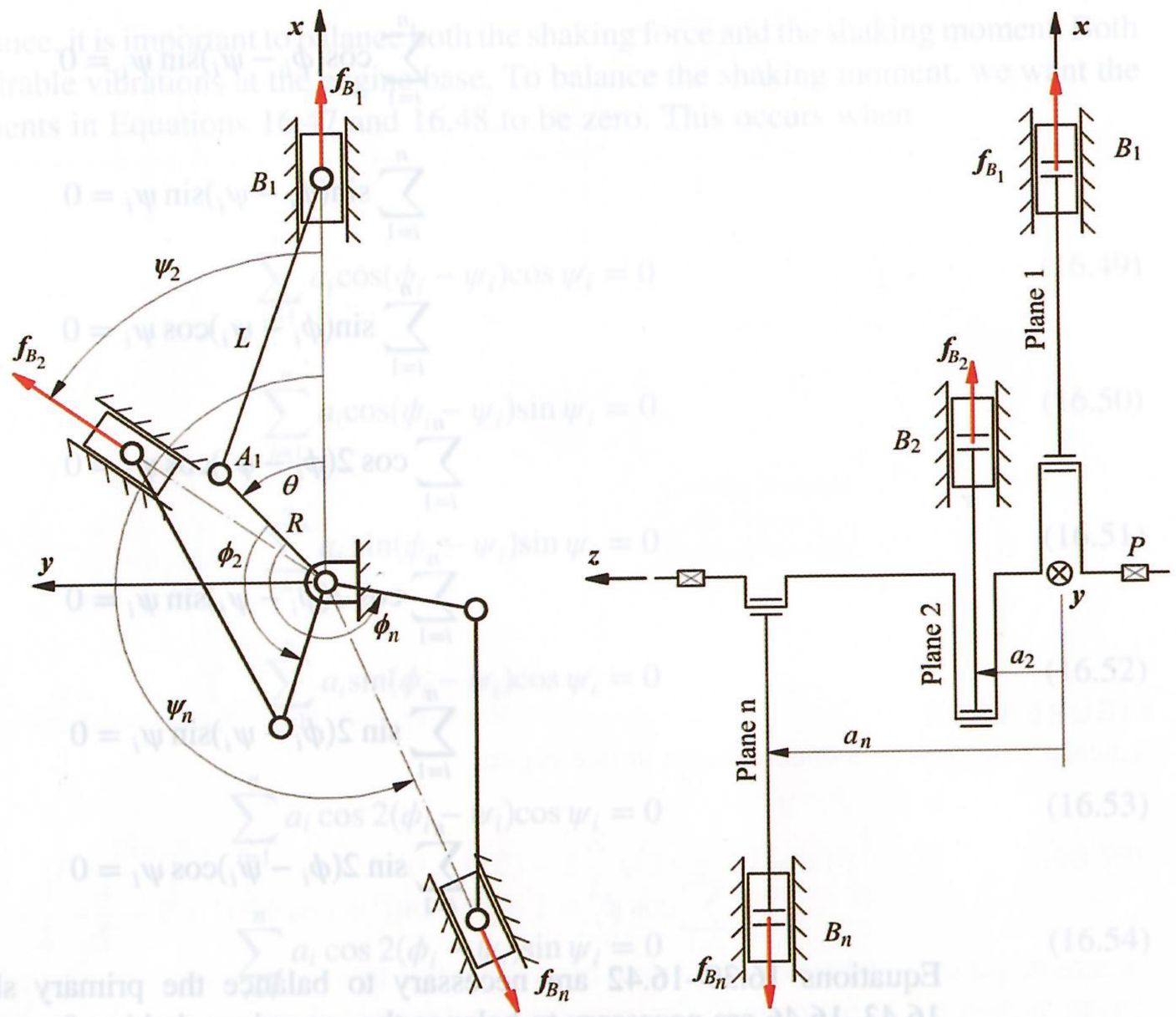


FIGURE 16.14
Schematic diagram of shaking forces in a multi-cylinder engine.

Equations 16.33 and 16.34 can be expanded to separate the fixed components from those dependent on θ using the following trigonometric identities

$$\cos(\theta + \phi_i - \psi_i) = \cos\theta\cos(\phi_i - \psi_i) - \sin\theta\sin(\phi_i - \psi_i) \tag{16.35}$$

and

$$\cos 2(\theta + \phi_i - \psi_i) = \cos 2\theta \cos 2(\phi_i - \psi_i) - \sin 2\theta \sin 2(\phi_i - \psi_i) \tag{16.36}$$

Then

$$f_x = \overline{m}_B R\omega^2 \left[\cos\theta \sum_{i=1}^{n} \cos(\phi_i - \psi_i)\cos\psi_i - \sin\theta \sum_{i=1}^{n} \sin(\phi_i - \psi_i)\cos\psi_i + \frac{R}{L}\cos 2\theta \sum_{i=1}^{n} \cos 2(\phi_i - \psi_i)\cos\psi_i - \frac{R}{L}\sin 2\theta \sum_{i=1}^{n} \sin 2(\phi_i - \psi_i)\cos\psi_i\right] \tag{16.37}$$

and